Project Report

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1 Introduction

Efficient management of urban traffic remains a critical global challenge, as congestion directly impacts economic stability, environmental sustainability, and public safety. **Traffic Signal Control** (TSC) is a fundamental component of urban mobility, requiring real-time decisions that must continuously balance operational efficiency with safety-critical priorities. While traditional rule-based heuristics and standard **Reinforcement Learning** (RL) methods have been widely adopted, they often struggle to generalize in complex or safety-critical scenarios, such as the management of emergency vehicles.

This project builds upon the architecture introduced by VLMLight, a novel TSC framework that integrates vision-language meta-control with dual-branch reasoning. The original VLMLight framework utilizes a Large Language Model (LLM) as a meta-controller to dynamically select between a fast RL policy for routine traffic and a deliberative reasoning branch for critical cases, such as the arrival of ambulances or fire trucks.

Following the work established in the VLMLight paper, this study specifically narrows its focus to the Reinforcement Learning (RL) component of the architecture, rather than the Large Language Model (LLM) reasoning branch.

The goal of this work is to refine and evaluate the routine control mechanism, which relies on a Transformer-based encoder to capture spatial dependencies and temporal dynamics among different traffic movements, having a Base Station (BS) that allocates resources to users moving in this traffic. Based on this foundation, two distinct strategies for Network Capacity Allocation at the intersection level were developed and implemented. These methods aim to optimize the distribution of capacity resources among competing traffic movements, ensuring that the decision-making process is responsive to real-time vehicle demand.

The methodology and implementation details for these two allocation modes, along with their comparative performance analysis, will be presented and discussed in the subsequent sections of this report.

2 Methodology

The main objective of this study is to optimize the allocation of network capacity at a fixed intersection. We define the problem as the dynamic distribution of a finite bandwidth resource (C_{max} = 50 Mbps) among competing traffic movements. To evaluate different approaches, we implemented and compared two primary strategies: an analytical baseline based on an optimization model and an RL based approach using as a skeleton the model we already have [1].

2.1 Strategy I: Analytical Direct Allocation

The Direct Strategy is the primary analytical approach explored. It is a deterministic mechanism that calculates the capacity distribution for each area of the BS by solving a continuous optimization problem in real-time. This strategy is built on the principle of Proportional Fairness (PF). Its goal is to maximize the aggregate network utility by balancing the total throughput with the individual needs of users in different cardinal sectors (North, South, East and West).

The Direct Strategy operates in a tightly coupled loop with the SUMO traffic simulator. Every simulation step follows four steps to transform raw traffic data into an optimal capacity mapping.

Phase 1 implies that the process begins by querying the TraCI (Traffic Control Interface) to obtain a high-resolution knowledge of the junction. The system identifies all unique lanes controlled by the junction ID (e.g., 'J1'), and for each lane, the system executes **getLastStepVehicleNumber**. This provides the exact instantaneous number of vehicles (d_{lane}) currently requesting network resources. In Phase 2 the code implements geometric mapping and source inversion. Because the BS allocates resources to 4 cardinal regions (North, South, East, West) rather than individual lanes, a spatial transformation is required.

The strategy calculates the movement vector $\vec{v} = (dx, dy)$ of each lane and then does source inversion logic. To properly serve the vehicles, the BS must activate the sector facing the incoming traffic. If a lane vector points North (positive Y), the vehicles are physically located in the South arm of the junction. The system applies a 180° rotation to the vector to map the demand to the correct position.

Lane-level counts are summed into four directional variables: d_{North}, d_{South}, d_{East}, d_{West}.

In Phase 3 there's mathematical solving. Once the aggregate demand d_i for each sector i is known, the system solves the Karush-Kuhn-Tucker (KKT) conditions for the following optimization problem:

$$\text{Maximize} \sum_{i \in \{N,S,E,W\}} d_i \ln\left(1 + \frac{c_i}{d_i}\right) \quad \text{subject to } \sum c_i = C_{max}$$

The solver implements the closed-form solution:

$$c_i^* = d_i \cdot \frac{C_{max}}{\sum_j d_j}$$

This ensures that the allocated capacity c_i is directly proportional to the demand d_i.

Last is Phase 4, where the system evaluates the performance represented by network utility. This metric represents the efficiency of the allocation, penalizing any sector that suffers from starvation (PF).

Interesting is explaining why PF was chosen as policy of allocation.

Max-Throughput is a purely utilitarian policy. It would allocate all bandwidth to the single busiest sector to maximize the raw sum of Mbps, leaving smaller groups with zero connection.

In contrast PF, the policy used in this strategy, uses a logarithmic utility function, so the system treats the cost of leaving a user without signal as a penalty. It forces the BS to distribute resources proportionally, ensuring that even a single vehicle in desolate regions has a minimum amount of capacity allocated.

It is important to underline how this strategy is created. It doesn't replace the existing Traffic Signal Control (TSC); instead, it leverages the pre-existing TSC framework to function effectively.

The Direct Strategy uses the same simulation environment and junction configurations established for the traffic lights. It is related to the existing **make_env** utilities to sync its decisions with the SUMO clock.

The BS uses the same lane IDs and junction definitions used by the TSC. This ensures that when the traffic light changes a phase, the BS immediately sees the resulting change in cars movements through the interface.

To maintain continuity with older versions of the framework and being able to make comparisons after, the strategy maintains a 5-frame moving average buffer.

This integration allows the BS to act as an overlay on top of the physical traffic infrastructure [1], adapting the allocation of capacity in real-time as the traffic lights move users through the intersection.

2.2 Strategy II: RL-Based Allocation (PPO with Transformer)

The RL approach to BS shifts the reasoning from a static mathematical solver to an adaptive agent. This strategy is centered on a Proximal Policy Optimization (PPO) algorithm, with usage of a Spatio-Temporal Transformer feature extractor.

Unlike the Direct Strategy, which only sees the current situation of traffic, the PPO agent processes a temporal window of the last 5 simulation steps. By using the mechanisms of the Transformer, the agent can identify complex correlations between different lanes and their evolution over time. This allows the agent to shift capacity allocation to areas that are about to experience high demand, rather than reacting only after the congestion has formed.

A fundamental feature of this strategy is that it operates as a secondary optimization layer on top of a pre-stabilized TSC system [1]. Before the BS

training begins, the junction is managed by a pre-trained PPO agent that has already optimized the traffic light phases to maximize vehicle throughput. The BS agent must therefore learn to allocate capacity within a highly dynamic environment where users are already being moved efficiently by the optimized TSC.

The study of this strategy has led me to analyze two possible ways in order to improve this method.

2.2.1 PPO using direct number of users

The first implementation of the PPO strategy is designed to bridge the gap between high-resolution traffic sensing and realistic hardware constraints. This model uses a detailed microscopic observation of 12 lanes but produces a macroscopic allocation across 4 cardinal sectors (North, South, East, West).

While we assume the standard data extraction pipeline and the Spatio-Temporal Transformer as the center of the system, it is important to highlight why this microscopic input is critical. By processing 12 individual lanes over a 5-step window, the Transformer extracts a 16-dimensional feature vector that contains informations such the number of cars and the state of the queue.

The output of this model is a 4-dimensional continuous action space. Each value represents a normalized weight assigned to one of the four cardinal sectors: North, South, East, and West. Unlike a direct model (Direct strategy), this policy treats each route arm of the junction as a single resource block. The capacity assigned to the "North" action is shared by all 3 lanes (Left, Through, Right) constituting that arm.

This 4-action output is a project choice in order to adapt to a real-world scenario where BS typically uses sectoral antennas that covers an entire street.

The agent is driven by a PF reward function evaluated using real-time demand. Even though the input is microscopic, the reward is synthesized at the sectoral level to match the output:

$$R = \sum_{i \in \{N,S,E,W\}} d_i \cdot \ln\left(1 + \frac{c_i}{d_i}\right)$$

The use of the logarithmic utility is vital here; it prevents the PPO from falling into a "Max-Throughput" trap where it would ignore quiet streets. Instead, the agent learns a policy that shifts the total capacity at the BS to the busiest regions, while maintaining a fair minimum for areas with lower users number. This strategy establishes a baseline for physically constrained optimization, providing a realistic benchmark to compare against the more precise, but more costly, individual lane strategy.

2.2.2 PPO Strategy: 12-Lane Focus

This is the second RL implementation thought about. In this configuration, the agent is liberated from the physical hardware constraints of 4-sector aggregation and is instead focused on a 12-dimensional continuous action space. This allows

the agent to allocate capacity with surgical precision to every individual lane of the junction.

The primary distinction of this strategy lies in its action space. While the first model serves cardinal arms (North, South, East, West), this strategy treats each lane as an independent resource requester. The agent observes 12 lanes (via the Transformer) and outputs 12 capacity weights. This symmetry allows the Spatio-Temporal Transformer to map its learned features directly to lane-specific needs without the less precise effect caused by sectoral aggregation.

The environment dynamically discovers the **unique_lanes** controlled by the traffic light at the start of the simulation **(traci.trafficlight.getControlledLanes)**. This ensures that the agent's 12 outputs are mapped correctly to the physical infrastructure of the selected scenario.

The strategy operates as a dual-agent system where the BS follows the lead of the TSC. The pre-trained Author's model (**tsc_model**) predicts the optimal traffic light phase, and the SUMO simulation advances, moving vehicles into their new positions. Notice that also the previous strategy described, has the same architecture.

The reward mechanism in this strategy is the most mathematically rigorous. Instead of rewarding the agent for satisfying a "North Sector" (which might contain both empty and full lanes) for example, it rewards the agent for the individual lane utility:

$$R = \sum_{i=1}^{12} d_i \cdot \ln\left(1 + \frac{c_i}{d_i}\right)$$

Because the agent can control c_i for each specific d_i, it learns to exploit the concavity of the logarithmic function. It realizes that moving capacity from an empty lane ($d = 0$) to a lane with 10 cars ($d = 10$) creates a massive jump in total network utility.

This strategy plays a vital role in this work as a benchmark. While it is weak from an Hardware point of view, it proves that the Transformer architecture is highly capable of identifying lane-specific traffic patterns. The PPO algorithm can successfully approximate the optimal solution when given a sufficiently precise action space.

We will understand in the following section the benefits and the limits of all these strategies.

3 Results

This study focuses its experimental validation on the Hongkong YMT (Yau Ma Tei) scenario. This junction was selected due to its high traffic density and complex asymmetrical flow patterns, which provide a rigorous stress test for dynamic capacity allocation. The junction is characterized by 12 controlled entry lanes, which were analyzed both individually and aggregated into four

cardinal directions (North, South, East, West) as described in the previous section.

Figure 1: YMT intersection

The evaluation was conducted by coupling SUMO (Simulation of Urban MObility) with a BS. The TSC was managed by a pre-trained PPO agent [1] to ensure realistic vehicle queuing and behaviors. To assess the robustness of the proposed Transformer-based BS, simulations were executed under three different total capacity limits (C_{max}): 50 Mbps, 20 Mbps and 10 Mbps. This multi-capacity approach allows for a sensitivity analysis of the agent's performance relative to resource scarcity.

The proposed RL agents were compared against two baseline strategies:

Uniform Allocation: A static policy that divides the total capacity equally among all parts, regardless of real-time demand.

Direct allocation: The direct strategy explained before. This policy assumes perfect knowledge of demand and serves as the theoretical limit for utility. Here we suppose to have all the informations we need about the channel.

The experimental results highlight a significant performance gap between the PPO with 12-lane focus and the one with 4-region output.

3.1 PPO Strategy: 12-Lane Focus results

As shown in the plots 2, the 12-lane agent achieves high adherence to the direct strategy curve.

At 10 Mbps, the model achieved its highest relative improvement (+3.75% gain vs Uniform). The agent's ability to precisely allocate capacity to specific lanes with active demand prevents the starvation of critical flows, a phenomenon that the Uniform policy cannot mitigate.

3.2 PPO using direct number of users results

The Analysis reveals the impact of hardware-realistic constraints.

This model showed a slight utility deficit (approx. -4.4% to -1%) compared to the Uniform baseline, as can be seen in 4. The visual analysis of the Dynamic Bandwidth Allocation plot indicates that when capacity is assigned to a cardinal

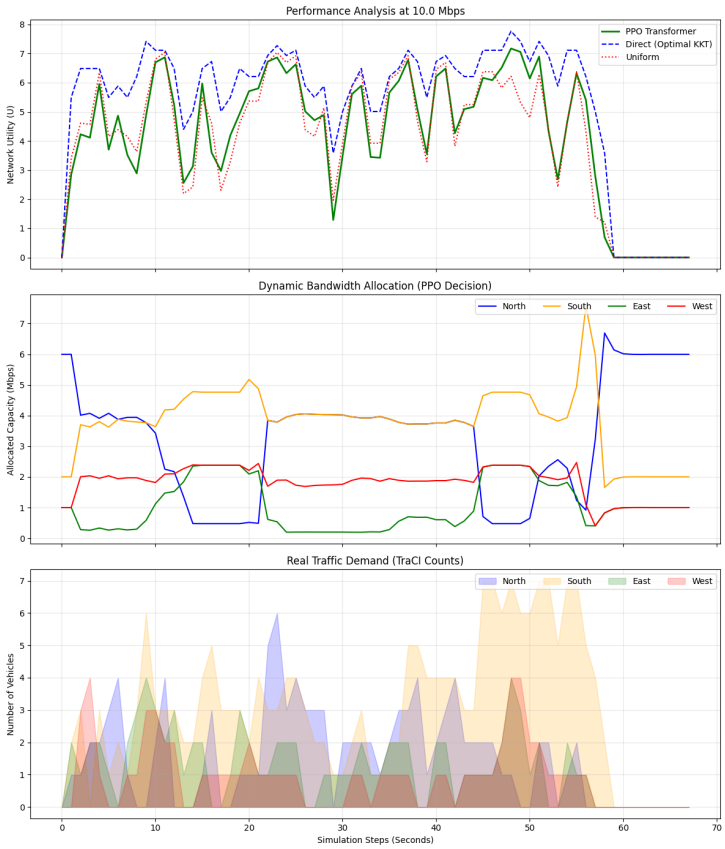

Figure 2: Performance Analyzis at 10 Mbps for 12-lane agent

Capacity	PPO_Avg	Uniform_Avg	Gain_%
50.0	10.687617	10.767093	-0.738141
20.0	6.688947	6.679782	0.137204
10.0	3.903853	3.761191	3.792989

Figure 3: Gain comparison: 12-lane PPO strategy vs uniform strategy

sector (e.g., North), it is distributed across all lanes in that sector. This leads to capacity dilution, where resources are wasted on lanes currently empty.

```
==================================================
 Capacity   PPO_Avg    Gain_%
     50.0 11.489777 -4.360700
     20.0  7.061166 -2.776911
     10.0  4.237065 -1.273178
==================================================
```

Figure 4: Gain comparison: 4-output PPO strategy vs uniform strategy

Interestingly, the gap between the agent and the Uniform baseline narrows as capacity decreases. At 10 Mbps, the agent nearly recovers to baseline levels, demonstrating that even a spatially-constrained RL agent becomes more valuable as the network approaches congestion.

By examining the Traffic Demand and capacity Allocation plots simultaneously, we observe that the 12-lane agent acts as a "mirror" to the traffic flow.

When a demand spike occurs (e.g., Northbound vehicles at step 55), the allocation plot shows an instantaneous and proportional increase in capacity for that specific slice. In contrast, the first model's allocation is more less precise, confirming that spatial precision is a more significant bottleneck for Network allocation efficiency than the intelligence of the RL algorithm itself.

4 Conclusion

This project investigated the application of a Spatio-Temporal Transformer based RL agent for dynamic network resources allocation in a complex urban environment. By evaluating the system in the Hongkong YMT scenario across different capacity regimes, several key conclusions can be drawn.

The experimental data reveal that the value of the RL-based BS is inversely proportional to the total available capacity. In high-capacity scenarios (50 Mbps), static policies like Uniform Allocation remain robust due to resource abundance. However, in constrained regimes (10 Mbps), the RL agent's ability to dynamically prioritize active flows resulted in a +3.75% utility gain, proving that AI-driven work is essential for preventing network collapse under congestion and critical situations.

A clear distinction emerged between the 12-lane Micro-Slicing and the 4-sector models. While the 12-lane model achieved high adaptation to the direct strategy optimum, the 4-sector model suffered from spatial aggregation penalties, where capacity was split across inactive lanes within the same cardinal direction. This highlights that the efficiency of network resource allocation depends not only by the algorithm's intelligence but also by the hardware's spatial precision.

Although the Direct Strategy provides a theoretical limit, the PPO agent remains a more reasonable and acceptable choice for real-world BS deployments.

Figure 5: Performance Analyzis at 10 Mbps for 4-areas agent

Unlike the direct approach, which requires continuous and perfect CSI and demand knowledge, often difficult to achieve, the PPO model learns to autonomously allocate capacity by extracting patterns from historical and local observations, being able to predict the most preferable allocation choice. This makes the RL-based approach significantly more robust to noise and delayed information in networks.

I propose in this part possible future works to continue this study. Several parts for future research remain to explore. A primary direction for future study involves the simulation of Emergency and Priority Scenarios. In cases of localized network surges (emergency vehicle preemption or mass-gathering events), resources cannot be divided equally. In such non-linear demand distributions, the Uniform policy is expected to fail significantly, allowing the PPO agent to demonstrate its full potential in ensuring capacity allocation.

A key priority for future validation is to extend the testing phase across a much larger number of episodes. By leveraging parallel processing (utilizing multiple CPU cores to run simultaneous SUMO instances), we can collect data from hundreds of different traffic seeds. This proposition comes from a study of the paper on which the work is based, where the training of the PPO is done on multiple processors [1]. This will allow for the calculation of confidence intervals, ensuring that the observed gains are statistically significant and not influenced by local traffic fluctuations.

Bibliography

[1] M. Wang, Y. Chen, A. Pang, Y. Cai, C. S. Chen, Y. KAN, and M. O. Pun, "VLMLight: Safety-Critical Traffic Signal Control via Vision-Language Meta-Control and Dual-Branch Reasoning Architecture," in *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL: `https://openreview.net/forum?id=08mjueZ0Iq`